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Optimized auxiliary classifier Wasserstein generative adversarial network fostered skin cancer classification from dermoscopic images

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Skin cancer is a disease that affects people of all ages. Automated diagnosis of skin cancer reduces the rate of death by detecting the disease at primary phase. Visual inspecting at the clinical inspection of skin lesion is one of the hard procedure because the similarity between the lesions exists. In this manuscript, Optimized Auxiliary Classifier Wasserstein Generative Adversarial Network fostered Skin Cancer Classification from Dermoscopic Images (OAC-WGAN-SCC-DI) is proposed. Initially, the input Skin dermoscopic images are engaged from the dataset of Skin Lesion Images for Melanoma Classification. The Dynamic Context-Sensitive Filter was used in removing noise and increasing the quality of Skin dermoscopic image. Next, these pre-processed images are given to Classic Semantic Segmentation Algorithm for segmenting ROI region.The segmented ROI region is given into Dual-Domain Feature Extraction for extracting Radiomic features such as Grayscale statistic features and Haralick Texture features. The extracted features are given into the Auxiliary Classifier Wasserstein Generative Adversarial Network (ACWGAN) which classifies the skin cancers, like Melanocytic nevus, Basal cell carcinoma, Actinic Keratosis, Benign keratosis, Dermatofibroma, Vascular lesion including Squamous cell carcinoma. In general, ACWGAN does not show any optimization adaption methods to determine the optimum parameterto offer accurate skin cancer classification. Artificial Humming Bird Optimization Algorithm is proposed in this manuscript to optimize ACWGAN classifier that classifies skin cancer precisely. The proposed OAC-WGAN-SCC-DI is implemented using MATLAB. To classify Skin cancer, performance metrics like precision, accuracy, F1-score, Recall (Sensitivity), Matthew's correlation coefficient, specificity, Jaccard co-efficient, Error rate, ROC, computational time are considered. Performance of the OAC-WGAN-SCC-DI approach attains 13.11%, 27.12% and 18.73% high specificity, 29.13%, 23.04% and 19.51% lower computation Time, 22.29%, 5.365%, 1.551% and 3.915% higher ROC and 28.65%, 3.98%, and 17.15% higher MCC compared with existing methods such as Skin cancer classification of Convolutional Neural Network with optimized squeeze Net by Bald Eagle Search optimization (DCNN-SCC-DI) and Skin cancer detection of Convolutional Neural Network using Gray Wolf Optimization (CNN-TL-SCC-DI), Hybrid convolutional neural networkswith SVM classifier for categorization of skin cancer (SVM-SCC-DI) respectively.

Keywords Skin lesion images, Dynamic context-sensitive filtering, Classic semantic segmentation algorithm, Dual-domain feature extraction, Auxiliary classifier Wasserstein generative adversarial network, Artificial humming bird optimization algorithm

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Skin cancer is one of the most dangerous types of cancer globally. The main forms include melanoma, basal cell carcinoma, squamous cell carcinoma, and intraepithelial carcinoma[1]. The human skin consists of three layers: the hypodermis, dermis, and epidermis[2]. The epidermis contains melanocytes, which can produce excessive melanin under certain conditions. Prolonged exposure to intense ultraviolet (UV) radiation from the sun can trigger this increased melanin production[3,4]. Melanoma, a particularly aggressive type of skin cancer, is caused by abnormal growth of melanocytes[5]. Early diagnosis of melanoma is crucial for effective treatment; if diagnosed early, the five-year survival rate is approximately 92%[6].

One of the significant challenges in melanoma diagnosis is the visual distinction between malignant and normal skin lesions[7]. This complexity makes the detection process difficult for trained professionals, as it relies heavily on visual examination alone[8,9]. Dermoscopy has emerged as a valuable imaging technique, providing a non-invasive method to photograph the skin's outer layer using a light source and immersion solution[10–13]. This method is widely used in dermatology and has been shown to increase the accuracy of malignant detection by around 50%[14–17]. However, human vision is inherently subjective and prone to errors, relying on the expertise of dermatologists for accurate diagnosis[18].

To improve melanoma identification, a Computer-Aided Detection (CAD) system is essential to assist professionals in the diagnostic process. Segmenting skin lesions is challenging due to variations in texture, size, color, and image quality. Poor contrast can hinder the differentiation of neighboring tissues, and extraneous details such as hair, blood vessels, and ruler lines complicate the segmentation process. Various methods have been employed to segment skin lesions, with Convolutional Neural Networks (CNNs) in deep learning achieving notable results. However, using lower resolution images to reduce computational demands can result in the loss of crucial visual information.

Existing research has explored optimization, segmentation, and deep learning classification approaches for skin cancer detection. Nevertheless, these methods often present issues such as complexity, time consumption, and high error rates. This study introduces the Optimized Auxiliary Classifier Wasserstein Generative Adversarial Network for Skin Cancer Classification Diagnosis (OAC-WGAN-SCC-DI) to improve classification accuracy.

The major contributions of this research work is summarized below:

- The Optimized Auxiliary Classifier Wasserstein Generative Adversarial Network (OAC-WGAN-SCC-DI) model is proposed for skin cancer classification using dermoscopic images.
- The model incorporates Dynamic Context-Sensitive Filtering (DCSF) to pre-process images and enhance quality for more accurate diagnosis.
- A Classic Semantic Segmentation Algorithm (CSSA) is employed to accurately isolate the Region of Interest (ROI), followed by Dual-Domain Feature Extraction (DDFE) for feature analysis.
- The Artificial Hummingbird Optimization Algorithm (AHBOA) is introduced to optimize the ACWGAN classifier for precise classification of skin cancer, achieving superior performance compared to existing methods

Remaining manuscript is arranged as follows: Part 2 reviews the Literature Survey, Part 3 depicts the proposed methodology, Part 4 exhibits the results, and Part 5 concludes this manuscript.

Literature survey

Various research works have been presented in the literature related to deep learning-based skin cancer classification. Some recent studies are reviewed below:

In 2021, Ali et al.[19] proposed an improved skin cancer categorization method utilizing a Deep Convolutional Neural Network (DCNN) combined with transfer learning techniques. Their approach involved three main steps: first, using a filter to reduce noise and artifacts; second, normalizing the input images and extracting features for accurate classification; and finally, applying data augmentation to increase the image count and enhance classification accuracy. The performance of the proposed DCNN was compared with various transfer learning models, including AlexNet, ResNet, VGG-16, and DenseNet, demonstrating high accuracy but a higher error rate.

In 2022, Kumar et al.[10] presented a skin cancer classification method for dermoscopy images using deep learning and transfer learning. They highlighted the critical issue of misdiagnosis in early-stage cancer, which can lead to severe consequences. Their study utilized Convolutional Neural Networks (CNN) and transfer learning to address these challenges. The results indicated a high level of accuracy, though the computation time was lengthy.

In 2023, Keerthana et al.[20] introduced a hybrid CNN with a Support Vector Machine (SVM) classifier for skin cancer categorization. They addressed the difficulty of automated skin lesion classification due to variations in lesion appearance. The proposed hybrid CNN methods concatenated features from two CNNs, which were then classified using an SVM. The method was validated against labels from professional dermatologists, achieving high sensitivity but lower accuracy.

In 2022, Rehman et al.[21] developed an automatic melanoma identification and classification system for dermoscopic images using a deep RetinaNet and a Conditional Random Field (CRF) approach. Their methodology included three main processes: image pre-processing, melanoma localization, and classification. The proposed technique was evaluated on the Pedro Hispano 2, International Skin Imaging Collaboration 2017, and ISIC 2018 datasets, yielding high specificity but low precision.

In 2021, Zakhem et al.[22] explored the role of dermatologists in developing artificial intelligence (AI) models for skin cancer assessment. They emphasized the importance of dermatologist involvement in integrating AI into medical practice. Their review, based on a PubMed search, highlighted advancements in AI for skin cancer detection, showing high Matthews Correlation Coefficient (MCC) but a low error rate.

In 2022, Attique Khan et al.[23] proposed a two-stream Deep Neural Network architecture for multi-class skin cancer classification. Their approach included a contrast enhancement strategy to improve images fed into a pre-trained DenseNet201 model, followed by a moth-flame optimization algorithm for feature optimization. Deep features were also extracted from a fine-tuned MobileNetV2 model. The method demonstrated high Jaccard Coefficient (JC) but low sensitivity.

In 2023, Alenezi et al.[24] introduced a multi-stage melanoma identification system based on a deep residual neural network and hyperparameter optimization. Their method featured realistic pre-processing techniques to remove noise, followed by feature extraction using a deep residual neural network. The Relief algorithm identified relevant features, which were classified using SVM. Bayesian optimization was applied to select optimal parameters, yielding high Receiver Operating Characteristic (ROC) values but low specificity.

In 2021, Singh et al.[25] developed a deep learning-based transfer learning architecture for the initial diagnosis and categorization of dermoscopic images of melanoma. Their Computer-Aided Detection for Melanoma Detection (CAD-MD) system integrated preprocessing, segmentation, feature extraction, and classification. The experiments utilized dermoscopic images from the PH2 and ISIC 2016 datasets, achieving high accuracy but requiring significant computation time.

Overall, these studies reflect ongoing efforts to enhance accuracy, efficiency, and clinical applicability in skin cancer classification using deep learning techniques.

Proposed methodology

In this section, OAC-WGAN-SCC-DIis discussed. This section presents the clear description about the research methodology used for the identifying the skin disease from the given dataset of Skin Lesion Images. The block diagram of OAC-WGAN-SCC-DI is represented by Fig. 1. Thus, the detailed description about OAC-WGAN-SCC-DI is given below,

Data acquisition

The dataset used for this classification task is the Skin Lesion Images dataset, which contains 9,810 dermoscopic images. It consists of 7,848 training images and 1,962 validation images, enabling comprehensive testing. This dataset is well-suited for skin cancer classification due to its balanced representation of different skin lesion types, including melanoma. Additionally, it has been carefully curated to reduce biases, making it an excellent resource for deep learning methods[26].

Preprocessing using dynamic context-sensitive filtering

Before training, the input dermoscopic images undergo a preprocessing step to remove noise and enhance their quality. This is achieved using Dynamic Context-Sensitive Filtering (DCSF)[27], a method that improves the extraction of image features by dynamically adjusting convolution kernels based on the image context.

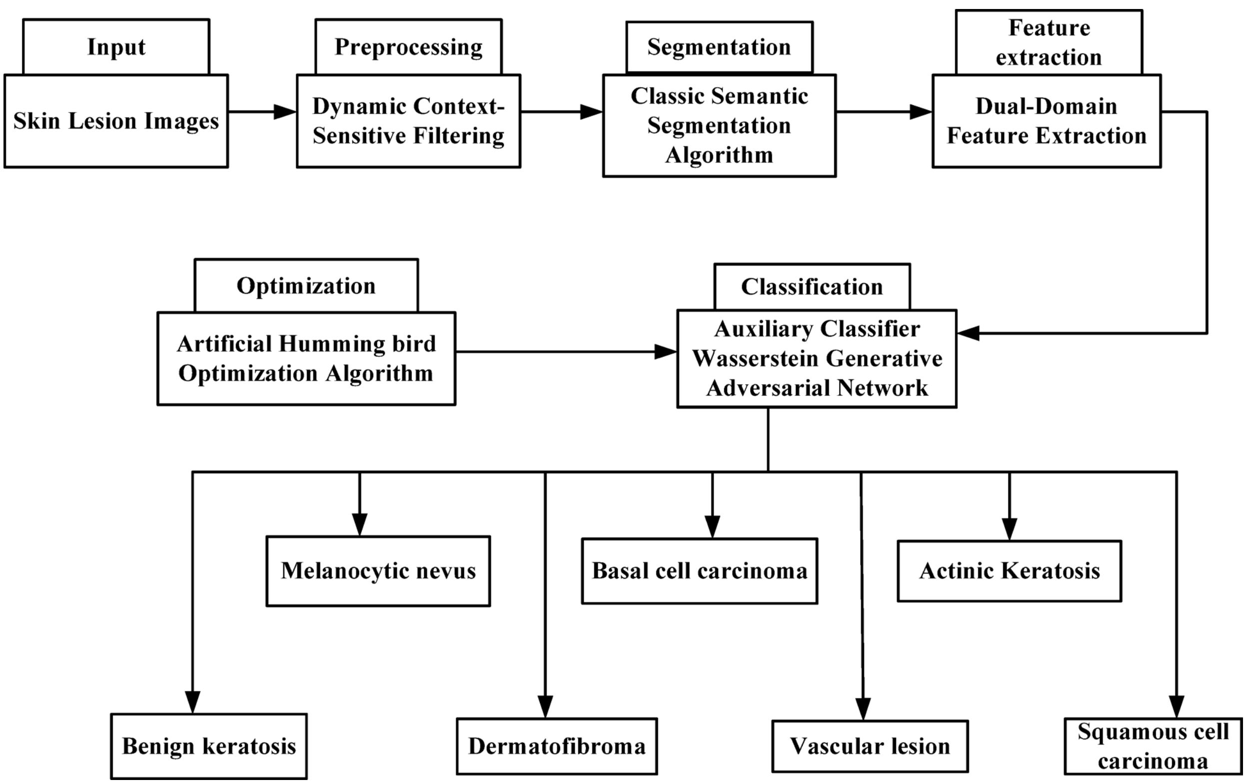

Fig. 1. Block diagram of the proposed OAC-WGAN-SCC-DI Skin Cancer Classification system.

The model predicts context-based weights that refine the image for better diagnosis. Key operations involve feature maps being processed through several convolutional layers, which reduces dimensionality and enhances the image by integrating attention mechanisms (Eqs. 1–6). This step improves the ability of the model to detect important features of skin lesions that are crucial for accurate classification.

As a result, Dynamic Convolution Kernels may extract rich contextual characteristics which was never constrained by the receptive field, offering complete advice to diagnose skin cancer accurately. DCFM takes 2 feature maps (C^{r-1}), (C^{r}) as the input. Every Filter-Generating Network forms a dynamic kernel $\tilde{S}_j^r$ is the subscript $j \in \{1,2,3\}$ added to differentiate among various kernel. Firstly, the network screates 3 feature map that are represented using an Eq. (1),

$$\alpha_j^r = conv_{1\times 1}(C^{r-1}) \tag{1}$$

where $conv_{1\times 1}$ signifies 1×1 convolution which decrease dimension of feature map to (C^{r-1}).The convolutional operations are represented using as Eq. (2),

$$\alpha_j^r = conv_{1\times 1}(C^{r}) \tag{2}$$

where (C^{r}) represents feature maps and $conv_{1\times 1}$ represents feature integration values and the values are represented using as Eq. (3),

$$\varphi_j^r = conv_{1\times 1}(C^{r}) \tag{3}$$

where φ_j^r represents dynamic kernel, the integration via an attention-guided weighted summation is represented using an Eq. (4),

$$u_j^r = C_{r_1}(AvgPooling(c_j^r)) \tag{4}$$

where, $AvgPooling$ shrinks input feature map from c_j^r and C_{r_1}, and condense it to the scalar values.

Then it is normalized with softmax function to form u_j^r. Final outcome may be represented using an Eq. (5)

$$\tilde{u}_j^r = \frac{d^{r_u^s}}{\sum_{i=1}^{3} c^r u^s} \tag{5}$$

where, $\tilde{u}_j^r$ represents scalar values $d^{r_u^s}$ indicates softmax functions and $\sum_{i=1}^{3} c^r u^s$ represents multi-scale features without distinctions the final output can be calculated using an Eq. (6)

$$Q^r = \sum_{i=1}^{3} u_j^r * D_i^r \tag{6}$$

where, Q^r represents convolutions with transforms and $\sum_{i=1}^{3} u_j^r * D_i^r$ indicates dynamic convolution kernels. Finally, the input image from Skin Lesion Images data set is efficiently preprocessed using Dynamic Context-Sensitive Filtering then the Skin lesion image is presented towards the segmentation phase.

Segmentation using classic semantic segmentation algorithm (CSSA)

After preprocessing, the images are segmented using a Classic Semantic Segmentation Algorithm (CSSA)[28], which identifies the boundaries of skin lesions. By properly identifying the lesion areas, this step ensures that the important regions of the image are isolated for feature extraction. Proper segmentation is crucial because it allows the model to focus on the lesion itself rather than irrelevant parts of the image, leading to better feature extraction and classification accuracy.

Feature extraction using dual-domain feature extraction (DDFE)

Following segmentation, the model extracts features from the image using Dual-Domain Feature Extraction (DDFE)[29]. This method captures both grayscale statistics (such as mean, skewness, and kurtosis) and texture features (such as contrast, homogeneity, and entropy). These features provide a detailed mathematical description of the skin lesions, which helps in distinguishing between different types of skin cancer. It helps to extract the Grayscale statistic features like standard deviation, mean, kurtosis, and skewness as well as Haralick Texture features, such as Contrast, Correlation, Energy, Homogeneity, Skewness, Variance, Standard deviation and Entropy. This technique has been utilized to extract features from input dermoscopic images.

Grayscale features are computed to describe the intensity distribution in the image (Eqs. 10–13). Texture features, on the other hand, capture the spatial arrangement of pixels, which helps in identifying patterns that are common in certain types of skin lesions (Eqs. 14–17). These features are then passed to the classification model for skin cancer diagnosis.

A quaternion formula, which is a development of the idea of complex numbers, is designated as the hyper complex input. It is given in Eq. (7)

$$l\left(x,y\right)=z+ue+vf+ws \tag{7}$$

where, e , f , s are the imaginary points, x , y are the quaternion variables, is the hyper complex input. This feature determines only the mean absolute difference each pixel and the average of the inner region and compare them to the same dimensions in the local background area. This is calculated using the Eq. (8),

$$KJ_{e,f}=\frac{K_{in}\left(e,f\right)}{x_{in}}-\frac{K_{out}\left(e,f\right)}{x_{out}} \tag{8}$$

where,$KJ_{(e,f)}$ represents the gray level value, $K_{in}\left(e,f\right)$ is the input neighborhood of the pixel e,f, $K_{out}\left(e,f\right)$ represents the output neighborhood of the pixel e,f. Hyper complex IR representation is used to combine the features. It is expressed in Eq. (9),

$$l\left(x,y\right)=h_1l_1+h_2l_2e+h_3l_3f+h_4l_4s \tag{9}$$

where, h_1,h_2,h_3,h_4 are the weight matrices, l_1,l_2,l_3,l_4 are the motion features. By this, the Grayscale statistic features and Haralick Texture features are extracted using DDFE model shown here,

Grayscale statistic features
The grayscale feature of the segmented images like standard deviation, mean, kurtosis, and skewness. They are discussed below,
Standard deviation (SD)
The standard deviation of the dermoscopic image feature can be expressed in Eq. (10)

$$SD=\sqrt{\frac{1}{A}\sum_{a=1}^{A}\left(B\left(a\right)-\beta\right)^2} \tag{10}$$

where β represents the mean.
Kurtosis (Kur)
The Kurtosis of dermoscopic image feature may be expressed by using Eq. (11)

$$Kur=\frac{C\left(B\left(a\right)-\beta\right)^4}{\lambda^4} \tag{11}$$

where, $C\left(\right)$ represents the expected values of the signal samples.
Skewness
Equation (12) used to calculate skewness of Grayscale statistic features.

$$S\left(i,a\right)=\frac{1}{A_s}\sum\nolimits_m\sum\nolimits_{n\in q}\left[u\left(m-1,n-r\right)-M\left(n,q\right)\right] \tag{12}$$

where, u represents Inverse Difference Normalized image, m represents Informative Measure of Correlations and q indicates spatial relationship of pixels.
Mean
Equation (13) is used to calculate the mean of Grayscale statistic features.

$$\beta=\frac{\sqrt{\sum_{k=1}^{m}p}}{M} \tag{13}$$

where β represents mean.

Haralick texture features
The Haralick Texture features of preprocessed images are discussed below,
Entropy
Equation (14) may be used to calculate entropy of statistic features of Grayscale.

$$F\left(L,s\right)=-\sum\nolimits_m\sum_{n=h}\left[u\left(m-h,n-x\right)\log^2\left(u\left(m-l,n-q\right)\right)\right] \tag{14}$$

where,$m-h$ represents spatial relationship of pixels and $n-q$ represents intensity values ,x represents coordinates of points.
Homogeneity
It is the similarity variation of the co-occurrence matrix in the image. Equation (15) represents calculating the homogeneity.

$$\mathrm{H} = \sum_{\mathrm{i=1}}^{\mathrm{N}} \sum_{\mathrm{k=1}}^{\mathrm{N}} \frac{\mathrm{GLCM\ (i,k)}}{1 + |\mathrm{i\ -\ k}|} \tag{15}$$

where $i - k$ represents element of the co-occurrence matrix.

Energy

Is the sum of the pixel point in an image and represented at Eq. (16)

$$E = \sum_{i=1}^{N} \sum_{k=1}^{N} GLCM\,(i,k)^2 \tag{16}$$

where, k and i represents variation in an images and N represents standard deviation of intensity.

Contrast

It is a measure of the existence of a variation in image pixels and it is represented in Eq. (17)

$$\sum_{i=1}^{N} \sum_{k=1}^{N} |i - 1|^2 \times GLCM\,(i,k) \tag{17}$$

where, i and k represents variation in the pixel and N indicates feature value. Finally, the input dermoscopic image is extracted using DDFE model. Then these extracted features are given into ACWGANto effectively classify the skin cancer diseases.

Skin cancer classification using auxiliary classifier Wasserstein generative adversarial network (ACWGAN)

This section focuses on **Skin cancer classification using auxiliary classifier Wasserstein generative adversarial network (ACWGAN)**[30]. ACWGAN is a powerful method typically used for image generation, but in this case, it is adapted for skin cancer classification.

ACWGAN works by generating images and assigning a class label to each generated image. The model is trained to distinguish between real and generated images, while also classifying each image into different skin cancer types. This dual objective—discriminating real from generated image and performing multi-class classification—makes ACWGAN particularly suited for this task. Equation (18) explains how the model determines the likelihood of the input being real or generated image. Equation (19) calculates the likelihood of the correct class for the input image. After training, the model uses its learned parameters to classify test images. The likelihood of the correct class for a test image is given by Eq. (20)**,** This step ensures that the model is able to classify images correctly even during testing.

The **Wasserstein GAN (WGAN)** component helps stabilize the training process by reducing problems like “mode collapse,” where the model fails to generate a variety of outputs. The stability of learning allows for more reliable tuning of hyperparameters (Eq. 21).

Eq. (21) outlines how the model calculates the minimum and maximum of the loss functions, aiming to strike a balance between generating realistic images and accurately classifying them

Optimization of ACWGAN

The model is further optimized through additional mathematical refinements (Eqs. 22–25), which involve using techniques like **RMSProp** instead of Adam for optimization, as it works better in certain cases. This improves the learning rate λ and ensures the model remains effective even under non-stationary conditions (where the data distribution changes over time).

Finally, while ACWGAN provides a robust method for skin cancer classification, additional optimization techniques can be applied to improve its performance further, specifically in terms of weight adjustments and workload predictions.

The likelihood of the correct source is given in Eq. (18)

$$Q_w = G\left[\log O\left(w = real|\, Z_{real}\right)\right] + G\left[\log O\left(w = fake|\, Z_{fake}\right)\right] \tag{18}$$

where, Q_w denotes correct source, Z_{real} is the real source, Z_{fake} is the fake source and O is the class label. The likelihood of the correct class is given in Eq. (19),

$$Q_D = G\left[\log O\left(D = d|\, Z_{real}\right)\right] + G\left[\log O\left(D = d|\, Z_{fake}\right)\right] \tag{19}$$

where, Q_D denotes the correct class, D is the train method. After that, parameter of the.

H method to generate test method $H\prime$ to categorize attest images and acquire the probability of accurate class. It is given in Eq. (20),

$$Q_{D\prime}\,(r)_{r=1}^{R} = G\left[\log O\left(D = d\prime\,(r)\,|Z_{real}\prime\right)_{r=1}^{R}\right] \tag{20}$$

where $Z_{real}\prime$ is the testing image. WGAN not only increase the stableness of learning and decrease issues including mode collapse and offer effective learning curves which is effective for debugging as well as hyper parameter search. It is given in Eq. (21),

$$\min_U \max_T G\left[H\left(Z_{real}\right)\right] - G\left[H\left(Z_{fake}\right)\right] \tag{21}$$

WGAN attains the stable method through determining minimum loss function of created method and discriminating the maximum loss function. Combined with ACWGANs multiple categorizations method, the specific practices contains 3 points, which is given in the Eq. (22),

$$u_b = \nabla_b \left[\frac{1}{r}\sum_{f=1}^{r} h_b \left(d, u_\varphi\left(Z_{real}\right) - \frac{1}{r}\sum_{f=1}^{r} h_b\left(d, u_\varphi\left(Z_{fake}\right)\right)\right)\right] \tag{22}$$

where, u_φ is expressed in Eq. (23),

$$u_\varphi = -\nabla_\varphi \frac{1}{r}\sum_{f=1}^{r} h_\varphi\left(d, u_\varphi\left(Z_{fake}\right)\right) \tag{23}$$

where, u_b represents sigmoid, r is the type of image, Z_{fake} is the fake source. Adam is substituted with RMS Prop because it is known to work effectively even for extremely non-stationary issues. It is given in Eq. (24)

$$b = b + \lambda.RMS\Pr op\left(\varphi, u_b\right) \tag{24}$$

where, φ is expressed in Eq. (25)

$$\varphi = \varphi - \lambda.RMS\Pr op\left(\varphi, u_\varphi\right) \tag{25}$$

where, λ is the learning rate. Finally, AC-WGAN accurately classifies the skin cancer. Generally AC-WGAN does not provide the optimization algorithms for calculating the ideal variables for confirming exact workload prediction. So the weight parameter a_u of AC-WGAN is essential to optimize by the optimization algorithm.

Optimization using artificial humming bird optimization algorithm (AHBOA)

The weight parameters of the proposed AC-WGAN are optimized using the Artificial Hummingbird Optimization Algorithm (AHBOA)[31], a bio-inspired approach based on the intelligent behaviors of hummingbirds. The key modules of AHBOA include food sources, hummingbirds, and the visit table.

Food sources

Each food source represents a possible solution to the optimization problem, where the quality of the nectar corresponds to the fitness value of that solution. Better solutions have a higher fitness value, which reflects the rate at which nectar is replenished.

Humming birds

Each hummingbird is assigned to a food source, and they share information about these sources with other hummingbirds. They can remember both visited and unvisited food sources, allowing them to explore new solutions effectively.

Visit table

This table tracks how frequently each food source is visited by hummingbirds. Sources with higher visitation levels are prioritized, guiding the hummingbirds towards the best food sources, or optimal solutions.

Stepwise procedure of AHBOA

Here, step by step procedure is defined to get ideal value of AC-WGAN based on AHBOA. Initially, AHBOA makes the equally distributing populace to optimize the optimum parameter a_u of AC-WGAN. The ideal solution is promoted utilizing AHBOA algorithm and the related flowchart is given in Fig. 2.

The step-by-step procedure of AHBOA for optimizing AC-WGAN is outlined below:

Step 1: Initialization.

A population of hummingbirds is initialized randomly, and each is assigned to a food source (Eq. 26). The bounds of the problem are set by the parameters:

$$a_u = low + e.\left(up - low\right)\ \ u = 1, \ldots n \tag{26}$$

where,low and up represents upper and lesser boundaries of d-dimensional issue, e denotes the random vectors at [0, 1], as well as a_u signifies location of the uth food source, which is solution of a given issue. It is given in Eq. (27)

$$VT_{u,v} = \left\{ \begin{array}{cc} 0 & if\ u \neq v \\ null & u \neq v \end{array} \right. \tag{27}$$

where for $u = v$, $VT_{u,v} = null$ denotes hummingbird took foodstuff at their particular source of food; for $u \neq v$, $VT_{u,v} = 0$ signifies vth food source is just visited through uth hummingbird at present iteration.

Step 2: Random generation.

The input parameters are generated randomly after initialization. The best solution is selected based on its fitness value, which is determined by specific hyperparameter conditions.

Step 3: Fitness function estimation.

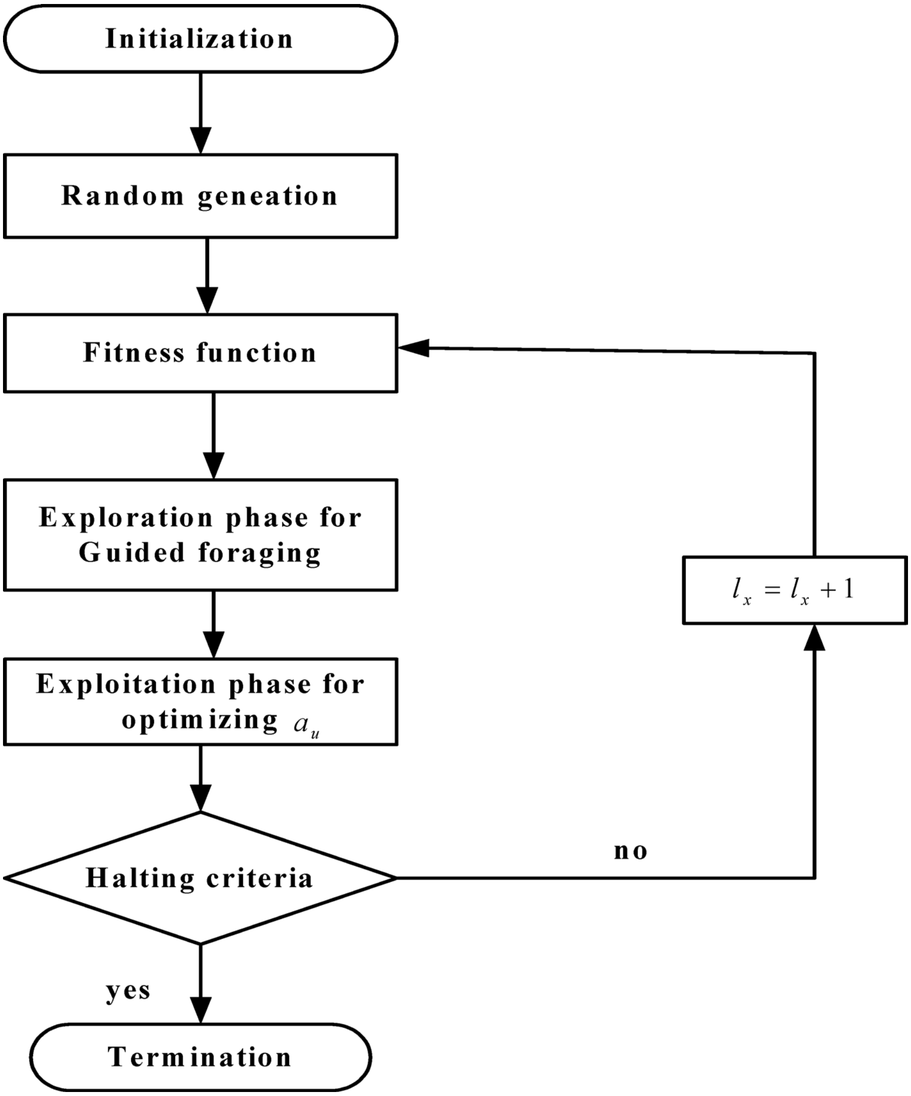

Fig. 2. Flowchart of AHBOA for optimizing ACWGANN parameter.

The fitness function evaluates how well a solution optimizes the weight parameter uauaua of the AC-WGAN classifier (Eq. 28):

$$fitness\ function = optimizing\ \ (a_u) \tag{28}$$

Step 4: Exploration phase for guided foraging.

In this phase, hummingbirds explore different food sources based on their nectar content and the time elapsed since the last visit (Eq. 29):

$$A^u = \left\{ \begin{array}{cc} 1 & if\ u = randu\left([1, a]\right) \\ 0 & else \end{array} \right. \tag{29}$$

where, $u = 1, 2, \ldots.a$. The diagonal flight is expressed in Eq. (30)

$$A^u = \left\{ \begin{array}{cc} 1 & if\ u = y\left(v\right) \\ 0 & else \end{array} \right. \tag{30}$$

where, y is the randperm. Along these flight capacities, a hummingbird visits the source of food it is seeking;as a result a candidate food source being acquired. The food source has been updated from the previous one depending target food source, which is chosen from overall exiting sources. It is given in Eq. (31),

$$m_u\left(h+1\right) = a_{u,tar}\left(h\right) + b.A.\left(a_u\left(h\right) - a_{u,tar}\left(h\right)\right) \tag{31}$$

where $a_u\left(h\right)$ denotes location of u^{th} source of food at the time h, $a_{u,tar}\left(h\right)$ signifies location of the goal food source that visit the u^{th} hummingbird, b indicates guided factor.

Step 5 Exploitation phase for optimizing a_u.

Once a food source is depleted, the hummingbird searches for a new one. This local search for candidate food sources is represented by the territorial factor ccc (Eq. 32):

$$a_u\left(h+1\right)=a_u\left(h\right)+c.A.a_u\left(h\right) \tag{32}$$

where, c is a territorial factor. It determines the extent to which the hummingbird explores new areas around its current position.

Step 6: Termination condition.

The process continues until a convergence criterion is met, typically defined by a halting condition like the number of iterations or a change in the fitness value. The halting criterion ensures that the optimization process does not continue indefinitely.

In this step, the weight parameter values of generator a_u from AC-WGAN is optimized by AHBOA Approach, will iterate the 3rdsteptill the halting criteria $l_x=l_x+1$ met**.** Then OAC-WGAN-SCC-DI classifies the skin cancer with better accuracy through reducing the computational time with error.

Result with discussion

The experimental outcome of suggested method is discussed in this section. The simulations performed in PC along Intel Core i5, 2.50 GHz CPU, 8 GB RAM, Windows 7. Then, proposed method is simulated utilizing MATLAB under performance metrics, like accuracy, precision, F1-score, Recall (Sensitivity), Matthew's correlation coefficient, specificity, Jaccard co-efficient, Error rate, RoC, computational time. The proposed OAC-WGAN-SCC-DI approach is implemented in MATLAB using Skin Lesion Images. The obtained outcome of the proposed OAC-WGAN-SCC-DI approach is analyzed with existing systems, DCNN-SCC-DI[20], CNN-TL-SCC-DI[21]and SVM-SCC-DI[22]respectively. The output of the proposed OAC-WGAN-SCC-DI method is given in Fig. 3.

Performance measures

This is a main task to select the ideal classifier. To analyze the performance, performance metrics, like accuracy, F1-score, precision, Recall (Sensitivity), Matthew's correlation coefficient, specificity, Jaccard co-efficient, Error rate, RoC and computational time are analyzed. To determine the performance metrics, the following confusion matrix is required.

- True positive (TP): correctly categorize a tumor image into tumor.
- True negative (TN): correctly categorize a tumor image into normal.
- False positive (FP): incorrectly categorize a normal image into tumor.
- False negative (FN): incorrect classification of a tumor image into tumor.

Accuracy

It isthe ratio of count of exact predictionwith total number of predictions made for a dataset. It is measured through Eq. (33),

$$Accuracy=\frac{(TP+TN)}{(TP+FP+TN+FN)} \tag{33}$$

Recall(R)

Recall isa metrics that calculates the predictions made by correct number of positive forecasts made by total positive predictions. It is measured by following Eq. (34),

$$\text{Recall}\left(R\right)=\frac{TP}{TP+FN} \tag{34}$$

Precision(P)

Precision isa metric which quantifies the count of correct positive prediction made.This is computed via following Eq. (35)

$$\Pr ecision=\frac{TP}{(TP+FP)} \tag{35}$$

F1-score

F-score is a metric used to analyze the performance of proposed DCNNtechnique. It is computed in Eq. (36),

$$F1\,Score=\frac{TP}{\left(TP+\frac{1}{2}\left[FP+FN\right]\right)} \tag{36}$$

ROC

An integrated measurement of a measurably effect or phenomena is the ROC. It is expressed in Eq. (37),

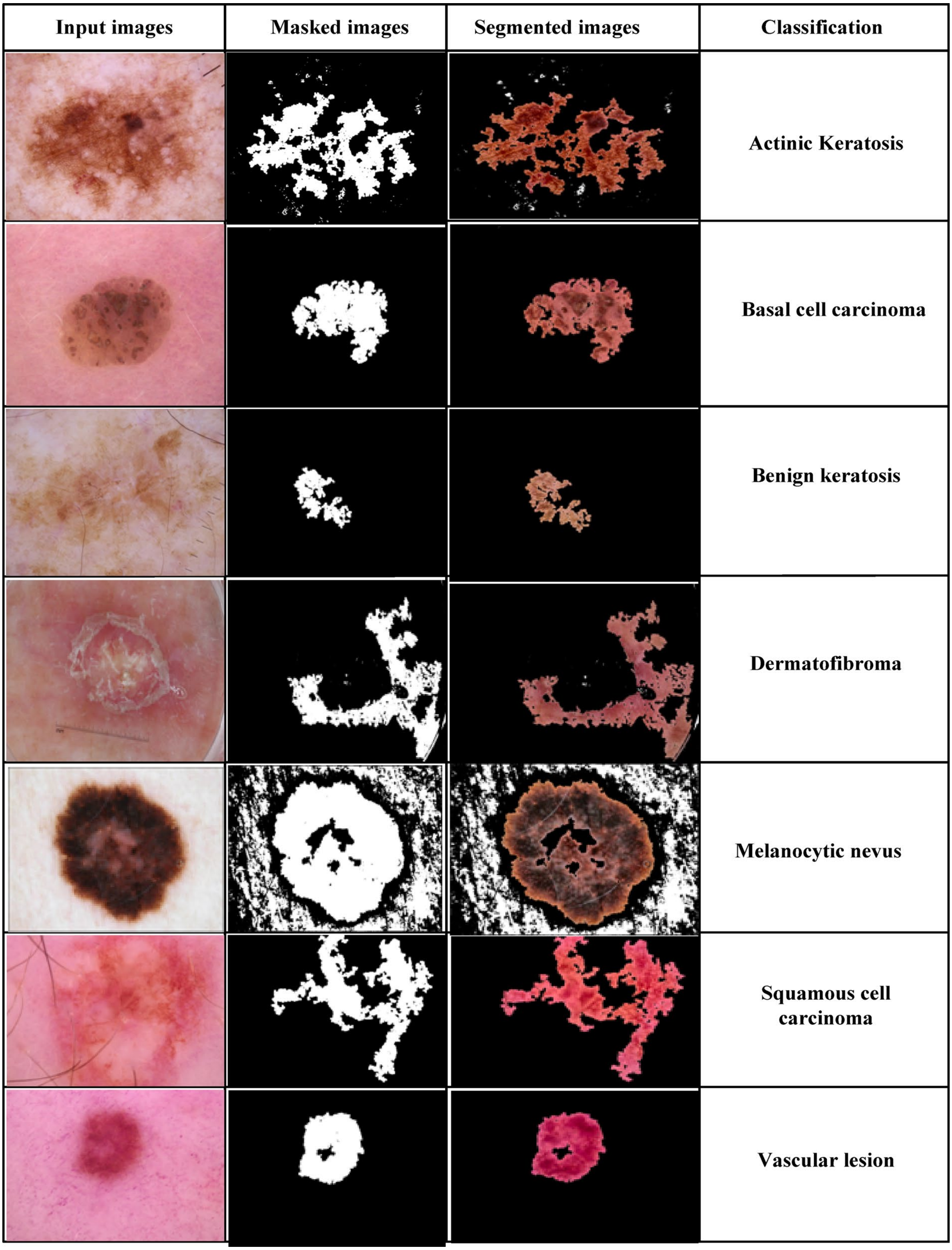

Fig. 3. Output result of proposed OAC-WGAN-SCC-DI methodology.

$$ROC = 0.5 \times \frac{TN}{FP + TN} + \frac{TP}{FN + TP} \tag{37}$$

Specificity
The percentage of true negatives that the method correctly identifies is called specificity. It is expressed in Eq. (38),

$$Specificity = \frac{TN}{TN + FP} \tag{38}$$

Matthew's correlation coefficient(MCC)
MCC measures the degree of correlation betwixt expected and actual values. It is expressed as Eq. (39),

$$MCC = \frac{TN \times TP - FN \times FP}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{39}$$

Jaccard Co-efficient (JC)
The Jaccard similarity is determined byseparating the count of observations in both sets through the count of observations in either set. It is represented by Eq. (40)

$$JC = \frac{TP}{TP + FP + FN} \tag{40}$$

Error rate
Error rate means a measure of degree of prediction error of a method made depending true method. It is scaled in Eq. (41)

$$Error\ Rate = 100 - Accuracy \tag{41}$$

Performance analysis

Figures 4, 5, 6, 7, 8, 9, 10, 11, 12 and 13 depicts simulation results of proposed OAC-WGAN-SCC-DI method. Then, the proposed OAC-WGAN-SCC-DI method is likened to existing DCNN-SCC-DI, CNN-TL-SCC-DI, SVM-SCC-DI methods.

Figure 4 shows Accuracy analysis. Here, proposed OAC-WGAN-SCC-DI method attains 21.51%, 52.38%, and 21.51% higher accuracy for Melanocytic nevus; 46.26%, 24.05%, 19.51% higher accuracy for Basal cell carcinoma; 33.78%, 39.43%, 52.30% higher accuracy for Actinic Keratosis; 34.72%, 19.75%, 31.08% higher accuracy for Benign keratosis; 52.38%, 36.31%, 39.13% higher accuracy for Dermatofibroma; 46.26%, 55.55%, 16.66% higher accuracy for Vascular lesion; 22.78%, 34.72%, 36.61% higher accuracy for Squamous cell carcinoma; compared with existing methods such as DCNN-SCC-DI, CNN-TL-SCC-DI, SVM-SCC-DI respectively.

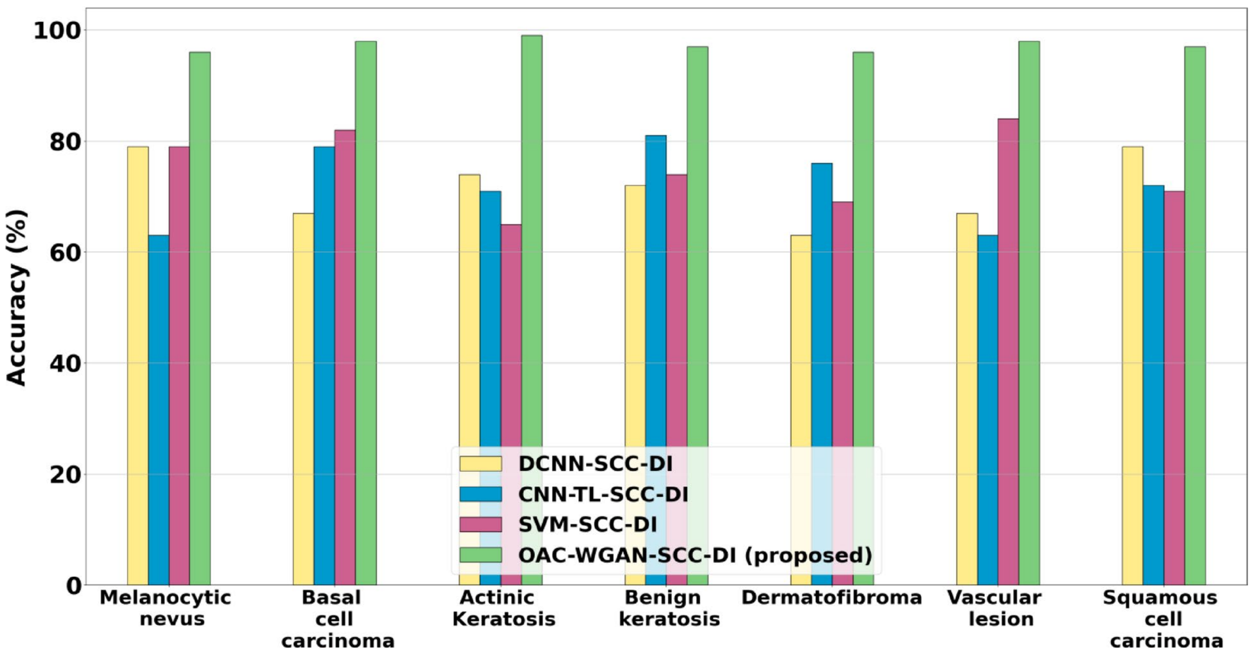

Fig. 4. Performance of accuracy analysis.

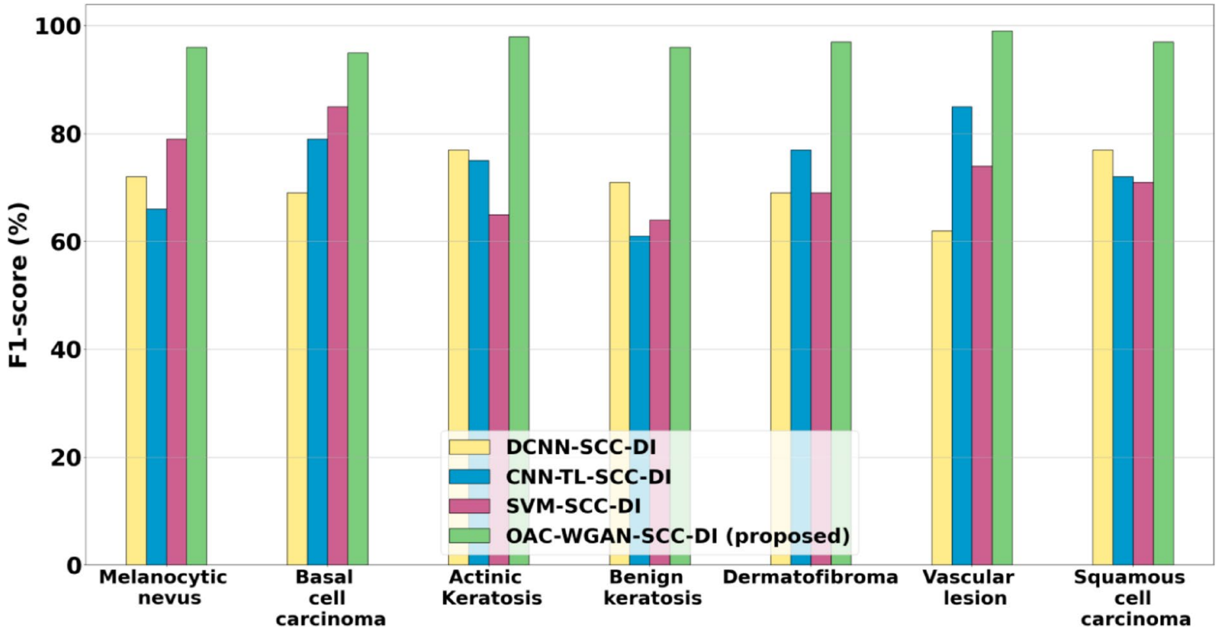

Fig. 5. Performance of F1-Scoreanalysis.

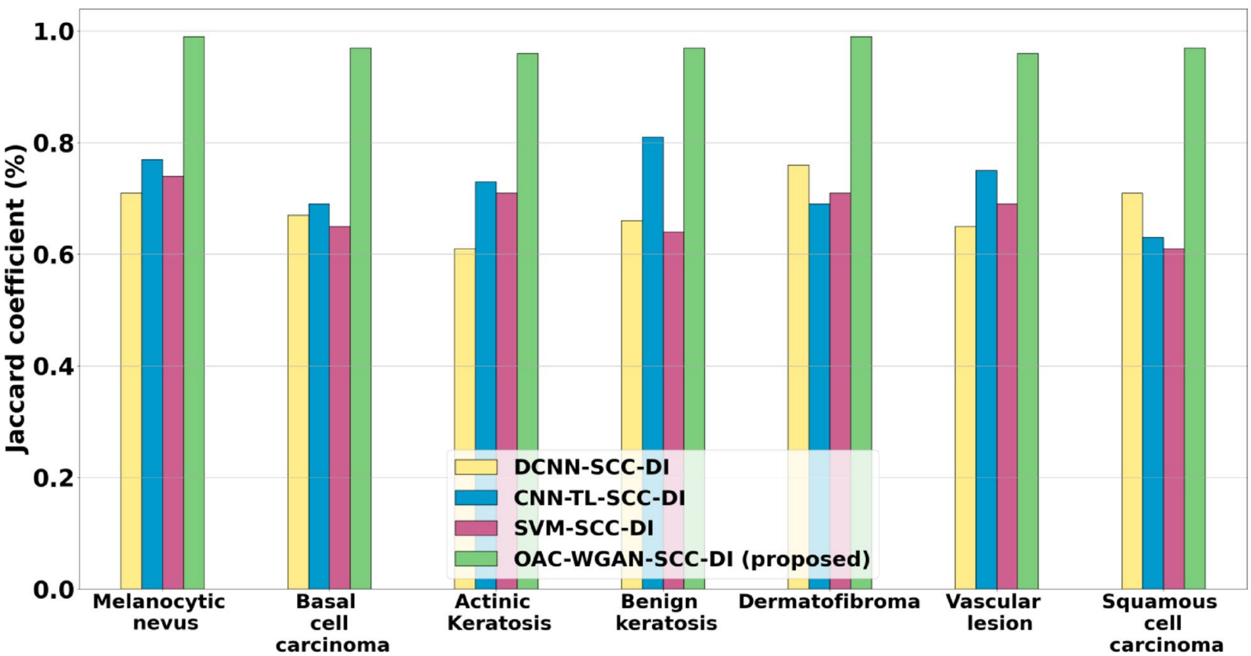

Fig. 6. Performance of JC analysis.

Figure 5 shows F1-Score analysis. Here, the proposed OAC-WGAN-SCC-DI method attains33.33%, 45.45%, and 21.51% higher F1-Score for Melanocytic nevus; 37.68%, 20.25%, 11.76% higher F1-Score for Basal cell carcinoma; 27.27%, 30.66%, 50.76% higher F1-Score for Actinic Keratosis; 35.21%, 57.37%, 50% higher F1-Score for Benign keratosis; 40.57%, 25.97%, 40.57% higher F1-Score for Dermatofibroma; 59.67%, 16.47%, 33.78% higher F1-Score for Vascular lesion; 25.97%, 34.72%, 36.61% higher F1-Score for Squamous cell carcinoma; comparing to the existing DCNN-SCC-DI, CNN-TL-SCC-DI, SVM-SCC-DI methods respectively.

Figure 6 shows JC analysis. Here, the proposed OAC-WGAN-SCC-DI method attains39.43%, 28.57%, and 33.78% higher JC for Melanocytic nevus; 44.77%, 40.57%, 49.23% higher JC for Basal cell carcinoma; 57.37%, 31.50%, 35.21% higher JC for Actinic Keratosis; 46.96%, 19.75%, 51.56% higher JC for Benign keratosis; 30.26%, 43.47%, 39.43% higher JC for Dermatofibroma; 47.69%, 28%, 39.13% higher JC for Vascular lesion; 36.61%,

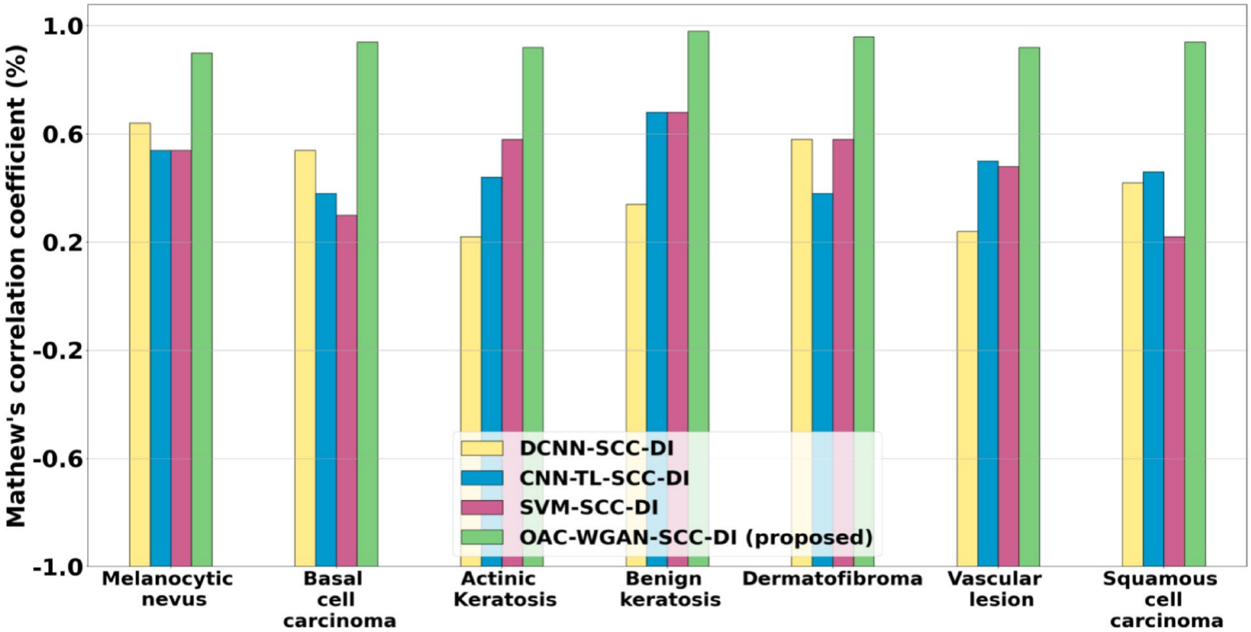

Fig. 7. Performance of MCC analysis.

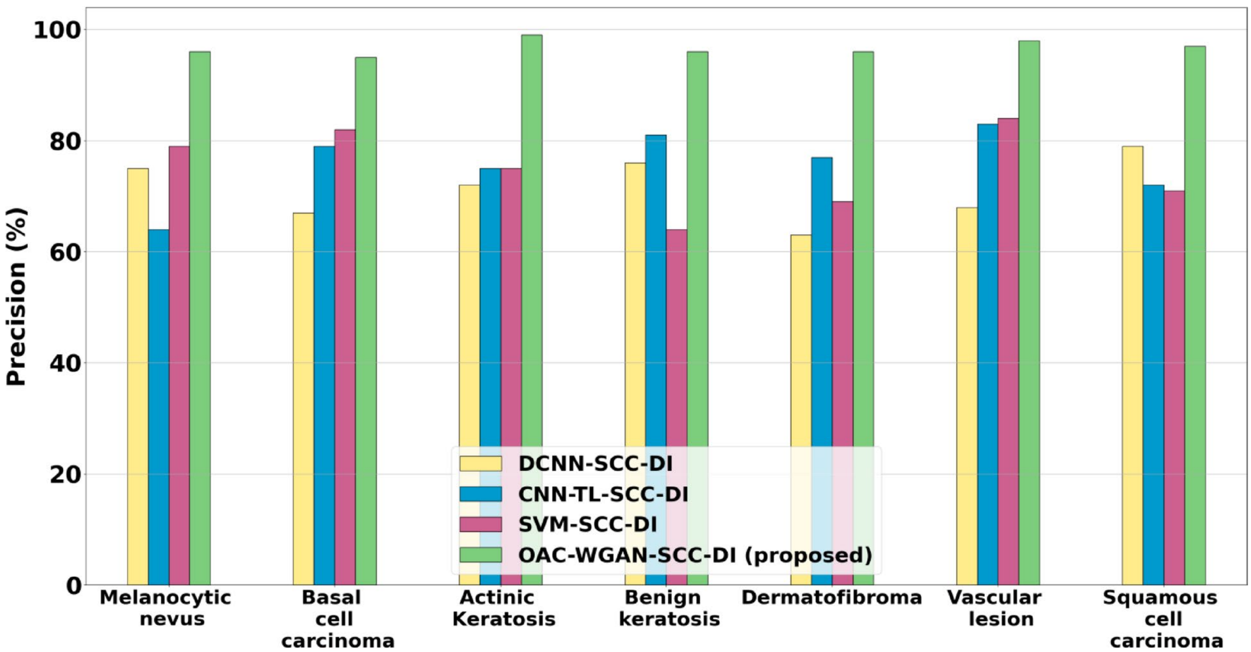

Fig. 8. Performance of precision analysis.

53.96%, 59.01% higher JC for Squamous cell carcinoma; compared with existing methods such as DCNN-SCC-DI, CNN-TL-SCC-DI, SVM-SCC-DI respectively.

Figure 7 shows MCC analysis. Here, proposed OAC-WGAN-SCC-DI method attains15.85%, 23.37%, and 23.37% higher MCC for Melanocytic nevus; 25.97%, 40.57%, 49.23% higher MCC for Basal cell carcinoma; 57.37%, 33.33%, 21.51% higher MCC for Actinic Keratosis; 47.76%, 17.85%, 17.85% higher MCC for Benign keratosis; 24.05%, 42.02%, 24.05% higher MCC for Dermatofibroma; 54.83%, 28%, 29.72% higher MCC for Vascular lesion; 36.61%, 32.87%, 59.01% higher MCC for Squamous cell carcinoma; compared with existing methods such as DCNN-SCC-DI, CNN-TL-SCC-DI, SVM-SCC-DI respectively.

Figure 8 shows Precision analysis. Here, proposed OAC-WGAN-SCC-DI method attains28%, 50%, and 21.51% higher Precision for Melanocytic nevus; 41.79%, 20.25%, 15.85% higher Precision for Basal cell carcinoma; 37.5%, 32%, 32% higher Precision for Actinic Keratosis; 26.31%, 18.51%, 39.13% higher Precision for

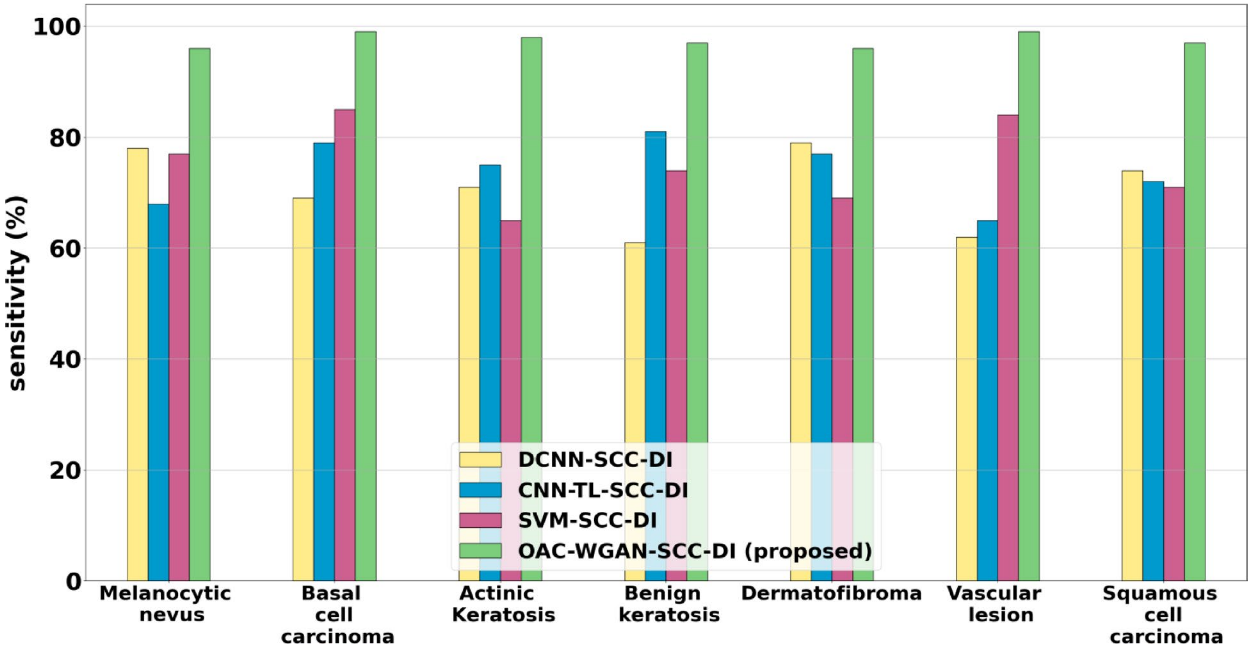

Fig. 9. Performance of sensitivity analysis.

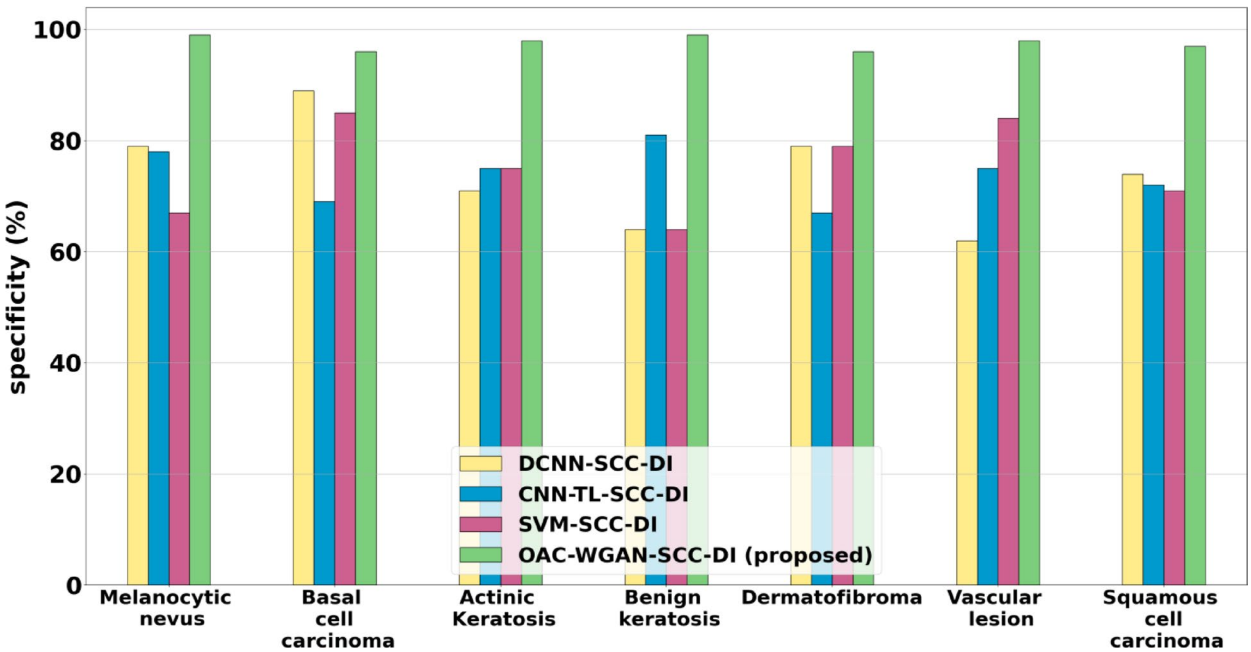

Fig. 10. Performance of specificity analysis.

Benign keratosis; 52.38%, 24.67%, 39.13% higher Precision for Dermatofibroma; 44.11%, 18.07%, 16.66% higher Precision for Vascular lesion; 22.78%, 34.72%, 36.61% higher Precision for Squamous cell carcinoma; compared with existing methods such as DCNN-SCC-DI, CNN-TL-SCC-DI, SVM-SCC-DI respectively.

Figure 9 shows Sensitivity analysis. Here, proposed OAC-WGAN-SCC-DI method attains23.07%, 41.17%, and 24.67% higher Sensitivity for Melanocytic nevus; 43.47%, 25.31%, 16.47% higher Sensitivity for Basal cell carcinoma; 38.02%, 30.66%, 50.76% higher Sensitivity for Actinic Keratosis; 59.01%, 17.95%, 31.08% higher Sensitivity for Benign keratosis; 21.51%, 24.67%, 39.13% higher Sensitivity for Dermatofibroma; 59.67%, 52.30%, 17.85% higher Sensitivity for Vascular lesion; 31.08%,34.72%, 36.61% higher Sensitivity for Squamous cell carcinoma; compared with existing methods such as DCNN-SCC-DI, CNN-TL-SCC-DI, SVM-SCC-DI respectively.

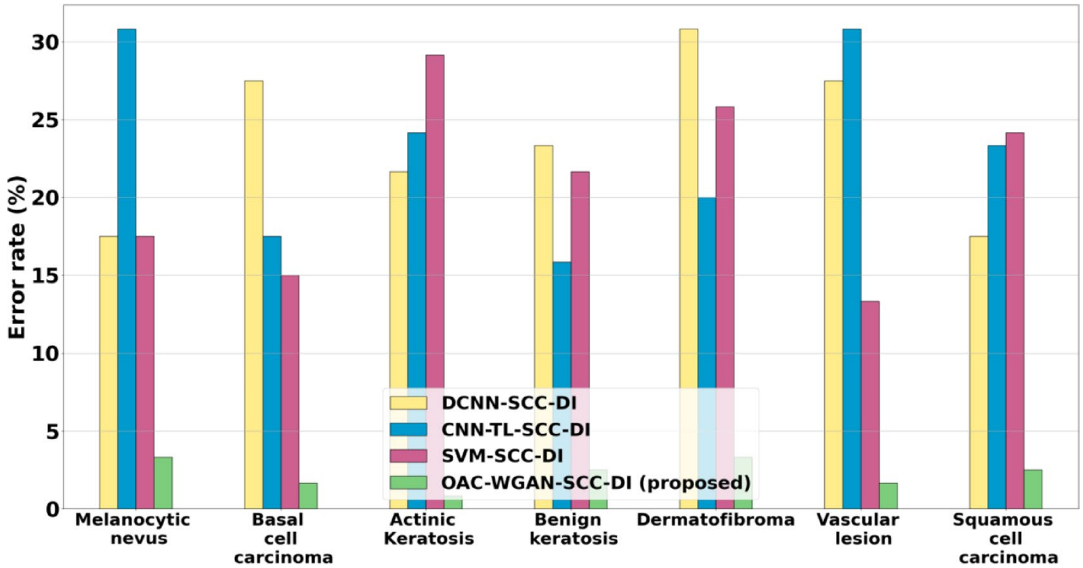

Fig. 11. Performance of Error rate analysis.

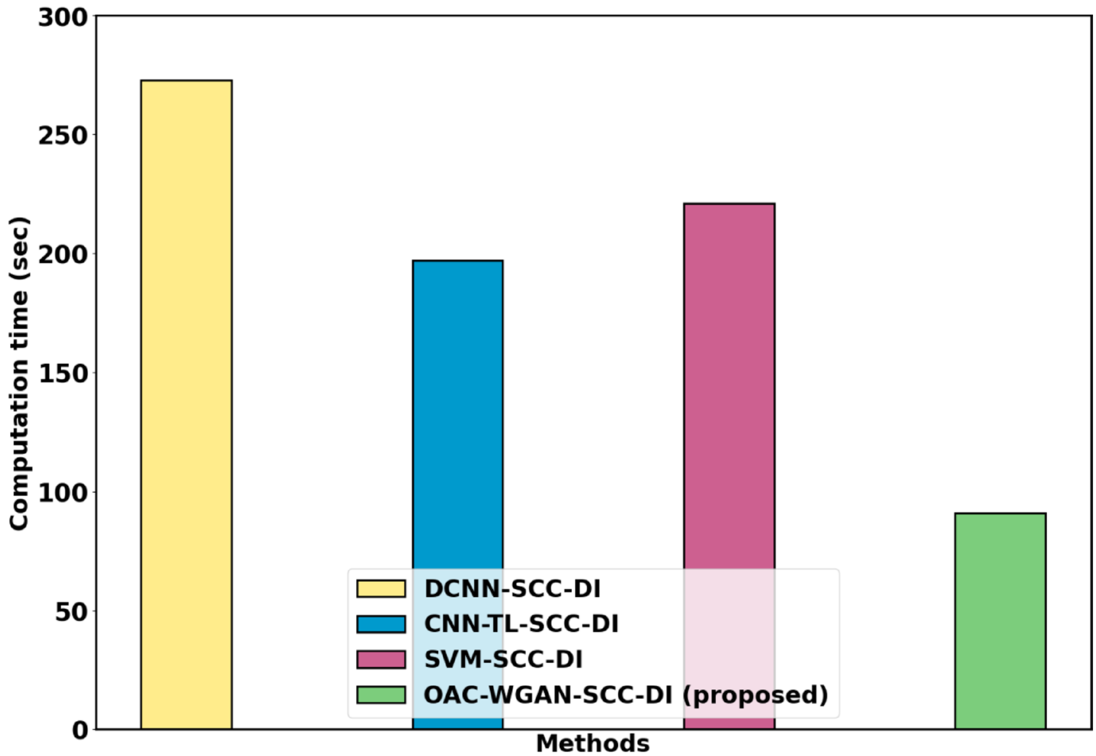

Fig. 12. Performance analysis of computation time.

Figure 10 shows Specificity analysis. Here, proposed OAC-WGAN-SCC-DI method attains25.31%, 26.92%, and 47.76% higher Specificity for Melanocytic nevus; 7.86%, 39.13%, 12.94% higher Specificity for Basal cell carcinoma; 38.02%, 30.66%, 54.68% higher Specificity for Actinic Keratosis; 54.68%, 22.22%, 54.68% higher Specificity for Benign keratosis; 21.51%, 43.28%, 21.51% higher Specificity for Dermatofibroma; 58.06%, 30.66%, 16.66% higher Specificity for Vascular lesion; 31.08%, 34.72%, 36.61% higher Specificity for Squamous cell carcinoma; compared with existing methods such as DCNN-SCC-DI, CNN-TL-SCC-DI, SVM-SCC-DI respectively.

Figure 11 shows Error rate analysis. Here, the proposed OAC-WGAN-SCC-DI method attains80.95%, 89.18%, and 80.95% lower error rate for Melanocytic nevus; 93.93%, 90.47%, 88.88% lower error rate for Basal cell carcinoma; 96.15%, 96.55%, 97.14% lower error rate for Actinic Keratosis; 89.28%, 84.21%, 88.46% lower

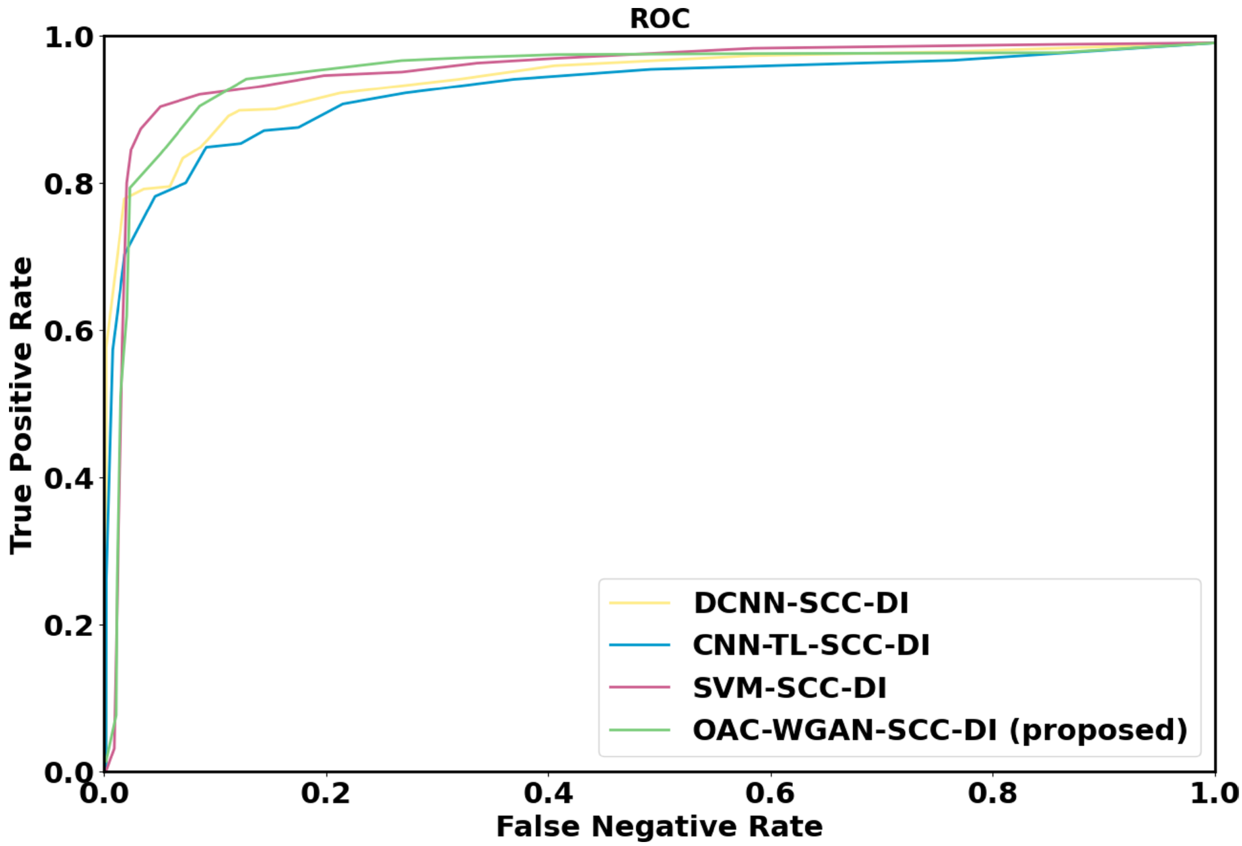

Fig. 13. Performance analysis of ROC.

error rate for Benign keratosis; 89.18%, 83.33%, 87.09% lower error rate for Dermatofibroma; 93.93%, 94.59%, 87.5% lower error rate for Vascular lesion; 85.71%, 89.28%, 89.65% lesser error rate for Squamous cell carcinoma; comparing to the existing DCNN-SCC-DI, CNN-TL-SCC-DI, SVM-SCC-DI methods respectively.

Figure 12 depicts Computation Timeanalysis. Here, proposed OAC-WGAN-SCC-DI method attains 66.66%, 53.80%, and 58.82% lower computation Time analyzed to the existing DCNN-SCC-DI, CNN-TL-SCC-DI, SVM-SCC-DI models respectively.

Figure 13 depicts the analysis of ROC. Here, the proposed OAC-WGAN-SCC-DI method attains 3.98%, 5.22%, 1.54% better AUC evaluated to the existing methods, such as DCNN-SCC-DI, CNN-TL-SCC-DI, and SVM-SCC-DI respectively.

Discussion

The OAC-WGAN-SCC-DI model introduces an innovative approach to skin cancer classification by leveraging a combination of advanced image processing techniques and optimization strategies. Central to this model is Dynamic Context-Sensitive Filtering (DCSF), which significantly enhances image quality by reducing noise. This enhancement facilitates the Classic Semantic Segmentation Algorithm (CSSA) in effectively isolating the Region of Interest (ROI) within dermoscopic images. Following this, the model employs Dual-Domain Feature Extraction (DDFE) to capture critical grayscale statistics and texture features, enabling precise differentiation among various types of skin lesions.

Experimental results validate that the proposed model achieves notable improvements in diagnostic metrics compared to existing methods, such as DCNN-SCC-DI, CNN-TL-SCC-DI, and SVM-SCC-DI. However, challenges remain, particularly regarding dataset diversity. The training and validation datasets may not adequately reflect the broad spectrum of skin types and conditions encountered in clinical practice, potentially impacting the model's generalizability.

Additionally, the model's higher computational requirements may present practical challenges for implementation, especially in resource-constrained healthcare settings. To address these limitations, future research should focus on expanding the training datasets to encompass a wider range of skin lesion types and patient demographics. This will enhance the model's adaptability across diverse clinical contexts. Moreover, efforts should be directed towards reducing the computational complexity of the model while preserving its accuracy. Investigating the model's robustness against adversarial attacks and developing strategies to mitigate overfitting are also essential areas for further exploration.

Conclusion

The OAC-WGAN-SCC-DI model signifies a substantial advancement in the automated classification of skin cancer from dermoscopic images. By employing a combination of innovative techniques, this model enhances the potential for accurate and efficient diagnosis in clinical settings. Its robust performance sets it apart from existing approaches, demonstrating its capability to improve early detection and treatment outcomes for skin cancer patients.

The integration of various advanced methods into a cohesive framework illustrates the model's applicability and effectiveness in real-world healthcare environments. Furthermore, the promising results indicate that ongoing development and refinement of this model could yield even greater benefits, leading to improved patient outcomes and broader implementation across diverse clinical settings. Overall, the OAC-WGAN-SCC-DI model provides a scalable solution that addresses existing challenges in skin cancer classification and has the potential to significantly impact patient care.

Data availability

The data used to support the findings of this study are included within the article.

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Declarations

Competing interests
The authors declare no competing interests.

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